**KT Document Out of Specification (OOS)**

1. **Purpose:**

The purpose of this document is to provide a comprehensive understanding of the OOS (Out of Specification) Management Software, including its features, workflow, and operational usage.

This document is intended to:

- Enable effective knowledge transfer to users, administrators, and stakeholders
- Ensure consistent, accurate, and compliant use of the system
- Support user training and onboarding activities
- Provide a reference for system functionality and process flow
- Facilitate smooth system handover and continuity of operations

The document acts as a structured guide to help users understand and operate the software in alignment with quality and regulatory requirements.

2. **Scope**

This document applies to the OOS (Out of Specification) Management Software used for handling, tracking, and managing OOS events within the organization.

The scope of this document includes:

- Recording and management of OOS events arising from laboratory analysis, manufacturing processes, or stability studies
- Execution of Phase I (Laboratory Investigation) and Phase II (Full Investigation)
- Documentation of root cause analysis and CAPA (Corrective and Preventive Actions)
- Review and approval workflows involving QC and QA personnel
- Tracking of OOS status from initiation to final closure
- Generation of reports for monitoring, trending, and compliance purposes

This document covers all users involved in the OOS process, including analysts, reviewers, QA personnel, and system administrators.

The system is intended to ensure proper documentation, traceability, and compliance with applicable regulatory and quality requirements.

3. **Responsibility**

This section defines the roles, responsibilities, and expected timelines for all stakeholders involved in the development and implementation of the OOS Management Software.

**3.1 Project Leader**

**Responsibilities:**

- Overall project planning and execution
- Requirement gathering and finalization
- Coordination between teams (Frontend, Backend, QA)
- Monitoring progress and resolving blockers
- Ensuring timely delivery and compliance with requirements
- Final approval of deliverables

### 3.2 Frontend Developer

Responsibilities:

- Design and development of user interface (UI)
- Implementation of forms (OOS entry, investigation, CAPA, etc.)
- Integration with backend APIs
- Ensuring responsive and user-friendly design
- Handling validations at UI level

### 3.3 Backend Developer

Responsibilities:

- Database design and schema creation
- Development of APIs and business logic
- Implementation of workflow logic (Phase I, Phase II, CAPA)
- Data validation and security controls
- Audit trail implementation

### 3.4 QA Tester

Responsibilities:

- Preparation of test cases and test scenarios
- Functional testing of all modules
- Validation of workflows and business rules
- Regression testing after bug fixes
- Ensuring system meets requirements

### 3.5 Knowledge Transfer (KT) Imparting Person

**Responsibilities:**

- Explain complete system requirements to development and QA teams
- Provide detailed workflow (OOS lifecycle, investigation stages, CAPA)
- Share data fields, validation rules, and business logic
- Clarify report requirements and expected outputs
- Resolve queries from developers and testers during development
- Ensure teams clearly understand system expectations before development

**4.0 Planned Timeline in Number of Days**

This timeline outlines the estimated duration for each stage of development, including knowledge transfer, development, testing, and bug resolution phases is as following:

| Stage | Duration (Days) |
|---|---|
| Knowledge Transfer | 5 |
| Development (FE + BE) | 20 |
| Integration | 5 |
| Internal Testing | 5 |
| Bug Fix (Internal) | 3 |
| QA Testing | 7 |
| Bug Fix (QA) | 3 |
| Final Testing | 5 |
| Bug Fix (Final) | 2 |
| **Total Duration** | **55 Days** |

**5.0 Team Composition**

| Name | Role |
|---|---|
| NKD | Team Lead |
| Rajkumar Patel | (Project Lead) Frontend |
| Rishi Gaware | Frontend |
| Sumit Bhalgamiya | Backend |
| Maan Desai | Backend |
| Sachin Kable | QA Tester |

**6.0 Definition**

**The following definitions are provided to ensure clarity and common understanding of terms used in the OOS Management Software.**

**Out of Specification (OOS)**
A test result that falls outside the predefined acceptance criteria or specification limits established for a product or material.

**Specification Limit**
The predefined acceptance criteria for a test parameter as defined in approved specifications or regulatory standards.

**Observed Result**
The actual value obtained during testing or analysis of a sample.

**OOS Event**
An occurrence where a test result is identified as Out of Specification and requires investigation.

**Phase I Investigation (Laboratory Investigation)**
The initial investigation conducted within the laboratory to determine if the OOS result is due to analytical error, instrument malfunction, or procedural issues.

**Phase II Investigation (Full Investigation)**
A detailed investigation conducted when no assignable cause is identified in Phase I, involving manufacturing processes, materials, methods, and environmental factors.

**Assignable Cause**
A clearly identified reason for the OOS result, such as analyst error, instrument failure, or deviation from procedure.

**Root Cause**
The fundamental reason responsible for the occurrence of the OOS event, identified after thorough investigation.

**CAPA (Corrective and Preventive Action)**
Actions taken to eliminate the root cause of an identified issue (Corrective) and to prevent recurrence (Preventive).

**Retest**
Re-analysis of the original sample to verify the initial OOS result under controlled conditions.

**Re-sampling**
Collection of a new sample from the same batch for re-analysis when justified.

**Investigation Status**
The current state of the OOS event (e.g., Open, Under Investigation, Closed).

**OOS Closure**
Final completion of the OOS process after investigation, root cause identification, CAPA implementation, and QA approval.

**Audit Trail**
A secure, time-stamped record of all actions performed within the system, including creation, modification, and approval of records.

**User Role**
Defined access level assigned to a user (e.g., Analyst, QA, Admin) determining system permissions.

**Workflow**
The predefined sequence of steps followed in the OOS process, from detection to closure.

**Deviation**
A departure from approved procedures or expected results, which may lead to an OOS event.

**Test Parameter**
A specific attribute or characteristic being tested (e.g., assay, dissolution, content uniformity).

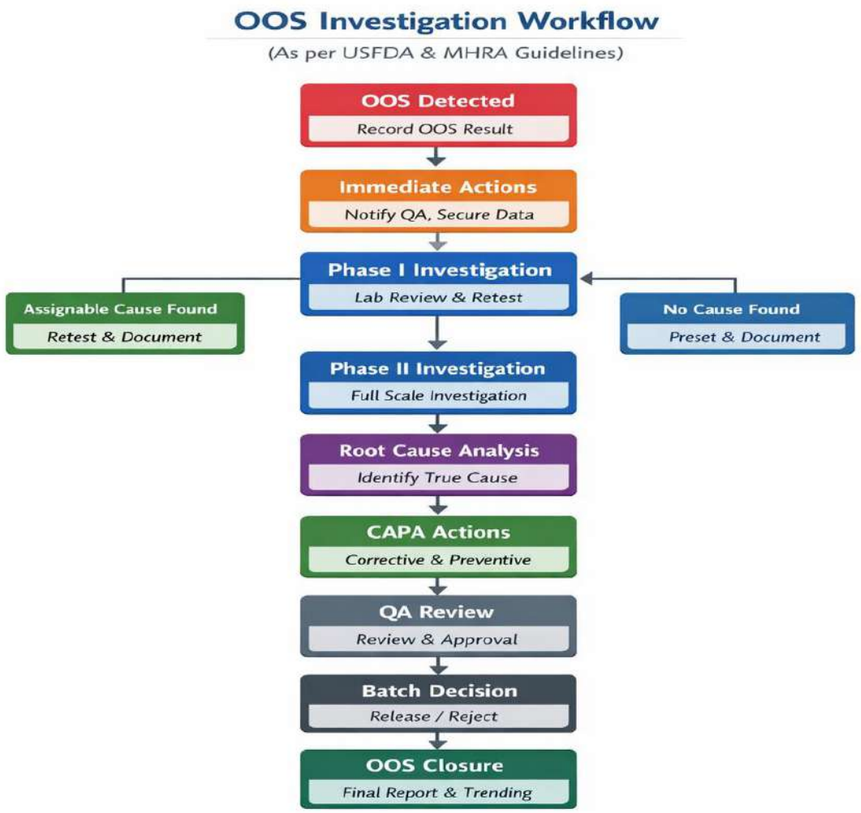
**Batch Number**
A unique identifier assigned to a specific quantity of material or product produced in a single process.

**Data Integrity**
The completeness, consistency, and accuracy of data throughout its lifecycle, ensuring compliance with regulatory requirements.

## 7.0 Regulatory Guideline Reference

| Sr. No. | Regulatory Authority | Guideline Name | Description | Access Link |
|---|---|---|---|---|
| 1 | US FDA (Food and Drug Administration) | Investigating Out-of-Specification (OOS) Test Results for Pharmaceutical Production | Provides detailed guidance on evaluation, investigation, retesting, and handling of OOS results in pharmaceutical manufacturing | Access FDA OOS Guidance |
| 2 | MHRA (UK) | Out of Specification (OOS) Investigations Guidance | Defines stepwise investigation approach including Phase I, II, and III investigations with decision trees and flowcharts | Access MHRA OOS Guidance |
| 3 | WHO (World Health Organization) | WHO Technical Report Series (TRS 986, Annex 2) | Provides GMP guidance including handling of OOS and investigation of abnormal results | General WHO GMP framework followed globally (IJMPR) |
| 4 | PIC/S (Pharmaceutical Inspection Co-operation Scheme) | PIC/S GMP Guide (PE 009) | Requires investigation of abnormal/OOS results, risk-based decisions, and proper documentation | Referenced as global GMP standard (IJMPR) |
| 5 | CDSCO / Schedule M (India) | Drugs and Cosmetics Act – Schedule M | Indian GMP requirements; expects alignment with WHO and USFDA practices for OOS handling | Indian regulatory framework aligned with global standards (IJMPR) |

**8.0 Process Flow for Out of Specification**



IF Phase_I_Result = "Lab Error"

→ Status = Invalid OOS → Close

ELSE

→ Move to Phase II

→ Perform Investigation

→ Identify Root Cause

→ Initiate CAPA (if required)

→ QA Final Approval

→ Close

**Process Flow: Out of Specification (OOS)**

**1. OOS Detection & Initiation**

- Analyst identifies result outside specification during testing
- OOS record is created in system

Capture Details:

• Product / Material

• Batch Number

• Test Name

• Specification Limit

• Observed Result

• Date of Analysis

• Instrument / Method used

Status: Initiated

**2. Phase I – Laboratory Investigation**

Objective: Identify any laboratory error

Performed By: Analyst / Lab Supervisor

Activities:

• Review raw data and calculations

• Check instrument calibration/status

• Verify method followed correctly

• Check sample preparation

• Analyst interview

Outcome:

- If Lab Error Found → Proceed to Invalid OOS
- If No Lab Error → Escalate to Phase II

Status:

• Phase I In Progress

• Phase I Completed

**3. Phase I Review & Decision (QA)**

QA reviews Phase I findings

Decision:

✓ If Assignable Cause (Lab Error):

- Mark OOS as Invalid
- Document justification
- Close OOS

✓ If No Assignable Cause:

- Approve escalation to Phase II

**4. Phase II – Full Investigation**

Objective: Identify root cause beyond lab

Performed By: Cross-functional team

Areas Covered:

• Manufacturing process

• Raw materials

• Equipment performance

• Environmental conditions

• Operator error

Activities:

• Batch record review

• Deviation check

• Trend analysis

• Re-testing (if applicable & justified)

Status:

• Phase II In Progress

• Phase II Completed

**5. Root Cause Identification**

- Identify scientifically justified root cause

Types:

• Manufacturing issue

• Material issue

• Method issue

• No root cause (rare, must justify)

**6. CAPA Initiation**

If required

Define:

• Corrective Action (fix current issue)

• Preventive Action (avoid recurrence)

Assign:

• Responsible person

• Target completion date

Status:

• CAPA Open

• CAPA In Progress

• CAPA Completed

**7. QA Final Review & Approval**

- QA verifies:

• Investigation completeness

• Root cause justification

• CAPA effectiveness

Decision:

• Approve → Close OOS

• Reject → Re-investigation required

**8. OOS Closure**

- Final status updated to Closed
- All records locked
- Audit trail maintained

Final Classification:

• Invalid OOS (Lab Error)

• Confirmed OOS

• Inconclusive

**9. Audit Trail (Throughout Process)**

- Every action recorded:
  - • User
  - • Timestamp
  - • Changes made

**10. Notifications & Escalations**

- Automatic alerts for:
  - • Pending approvals
  - • Delays (SLA breach)
  - • CAPA overdue

**Final Status Flow:**

**Initiated → Phase I → QA Review → Phase II → Root Cause → CAPA → QA Approval → Closed**

**9.0 Module Structure of Out of Specification Software**

- Module 1: Dashboard
- Module 2: Master Database
- Module 3: OOS Initiation
- Module 4: Phase I Investigation
- Module 5: Phase I Review
- Module 6: Phase I QA Review & Closure
- Module 6: Phase II Investigation
- Module 7: Phase II Review
- Module 8: QA Review & Closure
- Module 9: OOS Interview
- Module 9: Reports & Analytics
- Module 10: System Configuration

**10.0 Module and Page Wise Structure**

| Module Details | Page Details |
|---|---|
| Dashboard | Dashboard |
| Master Database | Product Master<br>Material Master<br>Line Master<br>Instrument Master<br>Test/Parameter Master |
| OOS Initiation | OOS Entry Form<br>OOS List View<br>OOS Detail View |
| Phase I investigation | Phase 1 Worklist<br>Investigation Approach<br>Investigation Approach Review<br>Investigation Approach Approval<br>Phase 1 Investigation Form<br>Phase 1 OOS List View |
| Phase I Review | |

**10.0 Dashboard Details**

| # | Parameter | Format | Rationale (Why Include) | Explanation (What It Means) | Visible To |
|---|---|---|---|---|---|
| 1 | Total OOS | KPI Card | Overall system load & trend | Total number of OOS records created in selected period | All |
| 2 | Open OOS | KPI Card | Identify pending workload | OOS not yet closed (any stage) | All |
| 3 | Closed OOS | KPI Card | Measure completion efficiency | OOS with final status = Closed | All |
| 4 | Invalid OOS | KPI Card | Track lab errors | OOS closed due to assignable lab error (Phase I) | QA, Admin |
| 5 | Confirmed OOS | KPI Card | Quality risk monitoring | OOS confirmed after full investigation | QA, Management |
| 6 | Inconclusive OOS | KPI Card | Identify weak investigations | OOS where no root cause found | QA, Management |
| 7 | Phase I In Progress | KPI / Bar | Track lab investigation workload | OOS currently under Phase I | Lab, QA |
| 8 | Pending QA Review (Phase I) | KPI | Identify approval delays | Phase I completed but awaiting QA decision | QA |
| 9 | Phase II In Progress | KPI / Bar | Monitor deeper investigations | OOS under full investigation | QA, Management |
| 10 | Pending QA Final Approval | KPI | Critical bottleneck indicator | OOS ready for final QA closure | QA |

| # | Parameter | Format | Rationale | Explanation | Visible To |
|---|---|---|---|---|---|
| 11 | Average Phase I Time | KPI / Line Chart | Efficiency of lab investigation | Avg time from OOS initiation → Phase I completion | QA, Management |
| 12 | Average Phase II Time | KPI / Line Chart | Investigation performance | Avg time for Phase II completion | QA, Management |
| 13 | Total OOS Cycle Time | KPI | Overall process efficiency | Time from initiation → closure | QA, Management |
| 14 | Overdue OOS | KPI (Red Highlight) | Compliance risk | OOS exceeding SLA timelines | All (especially QA) |
| 15 | Root Cause Distribution | Pie Chart | Identify recurring issues | % of OOS by root cause (method, material, process, etc.) | QA, Management |
| 16 | No Root Cause Cases | KPI | Audit concern indicator | Count of inconclusive investigations | QA |
| 17 | OOS Trend (Monthly) | Line Chart | Identify increase/decrease | Number of OOS over time | Management |
| 18 | OOS by Product | Bar Chart | Product risk identification | OOS count per product | QA, Management |
| 19 | OOS by Test | Bar Chart | Method reliability | OOS count per test parameter | QA |

**11.0 Report Details**

| # | Report Name | Details to be Included | Rationale (Why Required) |
|---|---|---|---|
| 1 | OOS Summary Report | OOS ID, Product, Batch, Status, Phase, Initiation Date, Closure Date | Provides overall visibility of all OOS records for tracking and audit |
| 2 | Open OOS Report | OOS ID, Current Stage, Responsible Person, Days Open | Helps identify pending investigations and workload |
| 3 | Closed OOS Report | OOS ID, Final Classification, Closure Date, Total Time | Demonstrates closure efficiency and audit readiness |
| 4 | Phase I Investigation Report | OOS ID, Lab Investigation Details, Findings, Analyst Comments | Ensures proper documentation of lab-level investigation |
| 5 | Phase II Investigation Report | OOS ID, Investigation Details, Departments Involved, Findings | Tracks detailed investigation beyond lab |
| 6 | Root Cause Analysis Report | OOS ID, Number of Root Cause, Root Cause Category | Helps identify recurring issues and process weaknesses |
| 7 | CAPA Report | OOS ID, CAPA ID, Action Details, Responsible Person, Due Date, Status | Tracks corrective and preventive actions for compliance |
| 8 | Overdue OOS Report | OOS ID, Stage, Responsible Person, Days Overdue | Identifies SLA breaches and compliance risks |
| 9 | OOS Trend Report | Month-wise OOS count, Product-wise trend, Instrument Wise Trend, Test wise trend, analyst wise trend, line wise trend | Helps management identify trends and recurring issues |
| 10 | OOS by Product Report | Product-wise OOS count and classification | Identifies high-risk products |
| 11 | OOS by Test / Parameter Report | Test-wise OOS distribution | Identifies problematic test methods |
| 12 | OOS by Root Cause Report | Root cause categories with counts | Helps in process improvement and CAPA planning |
| 13 | Audit Trail Report | User actions, timestamps, old vs new values | Mandatory for regulatory audits (21 CFR Part 11) |
| 14 | User Activity Report | User login, actions performed, approvals | Tracks user accountability and system usage |
| 15 | Approval History Report | OOS ID, Approver, Decision, Date, Comments | Ensures traceability of approval decisions |
| 16 | Inconclusive OOS Report | OOS with no root cause, justification | Highlights weak investigations (audit focus area) |
| 17 | Invalid OOS Report | OOS closed due to lab error | Tracks laboratory performance and errors |
| 18 | Turnaround Time (TAT) Report | Phase I time, Phase II time, total closure time | Measures efficiency of investigation process |
| 19 | Pending Approval Report | OOS awaiting QA or other approvals | Helps approvers prioritize actions |
| 20 | Reopened OOS Report | OOS reopened after closure with reason | Tracks quality issues and system gaps |